

let $S_m = \{s_{m,1}, s_{m,2}, \dots, s_{m,n}\}$ be the set of shares held by those contributors on that manuscript. By construction, contribution shares have the following properties:

- Shares for any contributor c (from authors A_m , peer reviewers P_m , and replicators R_m) on any manuscript m must be denoted as a real number between 0 and 1.

$$s_{m,c} \in [0, 1] \quad \forall c \in C_m = A_m \cup P_m \cup R_m \quad (1)$$

- The sum of contribution shares for all people involved on a paper for any manuscript is unity.

$$\sum_{k=1}^n s_{m,k} = 1 \quad (2)$$

- The value (academic capital) of a contributor's shares on a manuscript $\mathbf{x}_{m,c}$ is equal to the product of share of that manuscript and the (weighted) citations w of that manuscript.

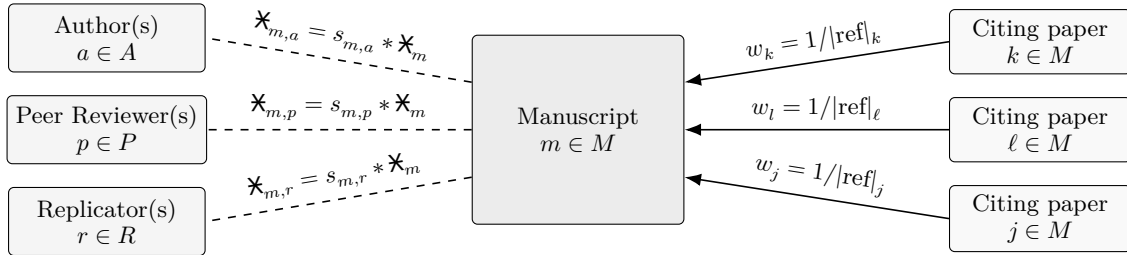
$$\mathbf{x}_{m,c} = s_{m,c} w_m \quad (3)$$

- All shares are fungible for the same manuscript.

$$\mathbf{x}_{m,c1} = \mathbf{x}_{m,c2} \iff s_{m,c1} = s_{m,c2} \quad (4)$$

Prior work on co-authorship credit allocation has proposed a range of weighting schemes, including equal fractional counting Egghe [2008], Perianes-Rodriguez et al. [2016], author-order-based heuristics Zhang [2009], Howard et al. [2007], and inferred contribution models based on citation structure Shen and Barabási [2014]. More recent work has emphasized the multidimensional and continuous nature of contributions Sauermann and Haeussler [2017], Allen et al. [2014], though most systems stop short of assigning quantitative shares.

The contribution share system in Liberata is closely related to fractional and contribution-based approaches in that it assigns continuous credit weights that sum to unity across contributors. However, it differs in a key aspect - *contribution shares are treated as explicit primitives rather than inferred quantities*. This allows the framework to generalize beyond authors to include other contributors such as peer reviewers and replicators. As such, the proposed system can be viewed as a natural generalization of existing co-authorship weighting schemes within a unified capital allocation framework.



Weighted citations from citing papers determine manuscript impact $\mathbf{x}_m = w_m = \sum 1/|\text{references}|$, which is then allocated across author, reviewer, and replicator roles in proportion to their shares in m .

Figure 1: Schematic of the Liberata framework. Contributor-role nodes hold manuscript shares $s_{m,c}$ $c \in C$ in the shares graph, while weighted citations define manuscript impact w_m in the references graph. Their composition yields manuscript-level academic capital allocations $\mathbf{x}_{m,c} = s_{m,c} \mathbf{x}_m$ across authors, peer reviewers, and replicators.

2.2 Weighted Citations & Correction Factors

2.2.1 Traditional Citations

Traditional citations, which Liberata refers to as absolute or unweighted citations u_m , counts the number of times a paper m appears in the references sections of later papers. This is a proxy measure for how "impactful" the paper is to future work. This is the primary way the impact of manuscripts are measured in

academia at present. Suppose M' represents the manuscripts published after m and the $\text{ref}(m', m)$ function returns 1 if m' has m in its references section, and 0 otherwise. Unweighted citations are then defined as:

$$u_m = \sum_{M'} \text{ref}(m', m) \quad (5)$$

This method of calculating impactfulness of a work is highly problematic for the following reasons.

- Citations do not indicate endorsement. Citations can indicate something is being refuted, criticized, or corrected. Yet regardless of whether a work is positively influential to another work, or completely negated by another work, that earlier work gets the same amount of credit.
- Citations are not normalized across disciplines. Different fields of academia differ substantially on typical size of references section, leading to high distortions in perceived impactfulness in different fields and especially for interdisciplinary scholarship.
- Citations are not normalized for the number of works in a reference section. This gives each paper published the ability to print unlimited credit into the universe and leads to the steady inflation of reference sections and frivolous citations.

While many of these caveats have been recognized in prior works, no concrete solution has emerged addressing all the shortcomings of impact quantification. Prior citation network based analytical works have explored different methods for weighting the edges to extract different metrics of interest and investigate network effects. For instance, bibliographic coupling Kessler [1963] and co-citation Small [1973] weightings construct similarity measures between documents based on shared references or joint citation patterns, and are widely used to uncover topical proximity and clustering structure in the literature. Derivatives of PageRank applied to citation graphs Page et al. [1999], Chen et al. [2007] instead emphasize global importance by recursively weighting citations according to the influence of the citing sources. Age-based weighting schemes Walker et al. [2007] incorporate temporal decay to privilege recent contributions or model citation dynamics, while field-normalized metrics Waltman et al. [2011] attempt to correct for systematic differences in citation practices across disciplines.

Despite their utility, these weighting schemes exhibit several limitations. Similarity-based methods such as bibliographic coupling and co-citation are inherently static and can overemphasize well-established areas, failing to capture emerging or interdisciplinary connections Porter and Rafols [2009], Wagner [2011]. PageRank-type approaches introduce rich-get-richer dynamics, which can amplify early advantages and entrench already prominent works or authors. Age-based weightings are sensitive to the choice of decay function and may undervalue foundational contributions Wang et al. [2013], Parolo [2015]. Field-normalized methods, while addressing cross-disciplinary heterogeneity, rely on often loosely-defined or externally imposed field classifications, which can be coarse, overlapping, or manipulable Waltman and van Eck [2013], Glänzel and Schubert [2010]. Moreover, the weighting methods explored in the prior works are susceptible to being used by citation cartels Fister [2016] to artificially bolster citation counts on their own articles. This problem has already been emerging in current systems of peer-reviewing Thurner and Hanel [2011], Szomszor [2020]. More broadly, many of these approaches lack explicit guardrails against cumulative advantage, leading to reinforcement of existing hierarchies and limiting their ability to fairly represent contribution in evolving scientific landscapes.

2.2.2 Liberata's Weighted Citations

The Liberata system introduces weighting mechanisms with corrections which, together, tackle all the listed problems of impact quantification. By default, Liberata system uses weighted citations w which normalizes the credit coming from each citation by the number of works cited in the referring paper, thus limiting the credit each paper can print from an uncapped citation number to 1 unit citation.

$$w_m = \sum_{k=1}^n \frac{1}{|\text{ref}|_k} \quad (6)$$

In addition to a normalization for reference count, there are two other optional correction factors that are applied by default in the Liberata system but can be toggled off or thought of as a variant of default metrics.

- Normalization for the publication rate of the academic field of the work. Suppose for a given academic field d_4 the average publication rate is $\rho(d_4)$ manuscripts per year per person. Then, we

normalize all weighted citations w coming from manuscripts $m \in M_{d_4}$ to

$$w' = \frac{w}{\rho(d_4)} \quad (7)$$

This implies that the Liberata system considers all academic fields equally productive on average, so that the amount of unit citations and $\star$ generated in each field per researcher is the same per unit time.

- Normalization for the author similarity of cited works. Suppose m' cites m . Let $\bar{v}_{m'}$ and $\bar{v}_m$ be vectors representing the share split amongst authors of the manuscript. The similarity is defined as:

$$\phi_{m',m} = \bar{v}_{m'} \cdot \bar{v}_m \quad (8)$$

The weighted citation going from m' to m is scaled by the author dissimilarity between the two works.

$$w'_{m',m} = (1 - \phi_{m',m})w_{m',m} \quad (9)$$

This correction factor is meant to nullify citation value of self accreditation.

2.3 Academic Capital

Academic capital ($\star$) is a core concept of the Liberata system. $\star$ is a measure of impact and contributions to academia and is defined over some set of papers M and shares held on M by people C .

$$\star_{M,C} = \sum_{m \in M, c \in C} s_{m,c} \cdot w_m \quad (10)$$

$\star$ is well defined over multiple situations. If we restrict C to a single contributor "c", then $\star_c$ represents c's career academic contributions across all academic fields. If we take C to be a set of individuals from a lab or institution, the impact and productivity of that lab or institution can be quantified in a way H-index Hirsch [2005] cannot generalize to (due to ill defined double/triple/... counting of citations). If we do not restrict C and instead restrict M to only be papers from certain fields or certain time periods, we can measure the volume of contributions within different fields of academic research or during certain time periods, or both. The versatility of defining subsets of M and C to examine allows for very simple yet powerful measures into the productivity and impact of different people, places, times, and research fields, from something as granular as the contributions of female authors from a particular institution during a period of time for a particular academic field, to something as broad as the total academic output of STEM fields globally per generation.

In addition to versatility, $\star$ is a superior metric to the h-index and citation count because it has a higher resolution while being a more accurate measure of individual contributions. This is possible because $\star$ has built in information not just of the ordinality of contributions, such as authorship position, but also of the distances in contribution, something traditional metrics do not have and cannot infer. Additionally, the normalizations and corrections of the citations allow for consistent interpretations across academic fields and time periods of the same numeric quantity.

2.4 Marketplaces

With Liberata's shares-based accreditation system, it is possible to recognize, in addition to the contributions of authors, those of other crucial players in the scientific process. Specifically, on the two marketplaces of the Liberata open access publishing platform, peer review and replication services, respectively, can be purchased by authors with contribution shares.

In the peer review marketplace, the authors can place a bid for peer reviewer services, specifying the number of peer reviews desired and percentage shares for each peer reviewer. To the authors, if the post-peer review manuscript m' , (on which the authors would have fewer shares but expect to accrue more lifetime citations), is likely to have higher academic capital than the pre-peer review manuscript m , then the authors are incentivized to go through peer review because the process has an expected net benefit on their conventional and new academic metrics (detailed in sections below). The equation below represents that condition for transaction.

$$\begin{aligned} \mathbb{E}[\star_{m',A}(s_{m',A}, \mathbb{E}[w_{m'}])] &> \mathbb{E}[\star_{m,A}(s_{m,A}, \mathbb{E}[w_m])] \\ \text{where } s_{m',A} &< s_{m,A} \text{ and } \mathbb{E}[w_{m'}] > \mathbb{E}[w_m] \end{aligned} \quad (11)$$

Alternatively, because $\mathbb{X}$ is proportional to s and w , this condition can also be expressed as:

$$E[w_{m'}] > \frac{s_{m, A}}{s_{m', A}} E[w_m]$$

Conversely, if the above were not true, the authors would not be incentivized to go through the peer review process and could leave their manuscript on Liberata as an unreviewed manuscript which is still readable to all and equivalent to the preprint state of manuscripts today on open access platforms such as arXiv.

For the peer reviewer p , it is worth taking $s_{m', p}$ to do this peer review if the time taken for the peer review t_p is less than $s_{m', p}$ of what it would take to research and author a similar quality work (measured by $E[\mathbb{X}]$) by themselves t_a .

$$t_p < t_a \cdot s_{m', p}$$

If both of these conditions are met, a transaction can be facilitated by a platform such as Liberata to allow for peer review to happen where both sides are incentivized and interest aligned to do good faith quality control. Unlike in other peer review systems, here the peer reviewers could expect to see greater returns if the work they reviewed actually ended up being better written or more accurate .

In the replication marketplace, the authors can place a bid for replication services, specifying the number of replicators desired, the percentage shares given to each successful replicator, and the time allotted for replication. A similar logic exists for authors on this marketplace as compared to the peer review marketplace, where the authors are comparing the expected academic capital of the fewer author shares on the post-replication paper m'' to the greater author shares on the pre-replication (post-peer review) paper m' . If the following is true, it is worth it for the authors collectively to pursue replication services.

$$E[w_{m''}] > \frac{s_{m', A}}{s_{m'', A}} E[w_{m'}]$$

Simultaneously, the replicator is comparing the time it would take to do the replication study on m' compared with the opportunity cost of working on their own paper. The condition they need satisfied for the replication to be worth doing is.

$$t_r < t_a \cdot s_{m'', p}$$

Like with the peer review marketplace, if both of these conditions are met, a transaction can occur, allowing for a replication study to happen where the replicator is incentivized to do good faith quality control and is interest aligned with the author. Here, replicators could expect to see greater returns on their contribution shares if the work they replicated actually ended up with a revised, more accurate conclusion, or was replicated, verified, and thus more trustworthy.

2.5 Academic Graphs

Graphs are a foundational tool in scientometrics, used to represent and analyze the structure of scholarly communication Fortunato [2018]. Traditionally, the academic ecosystem is typically modeled as a directed graph in which nodes correspond to entities such as manuscripts, authors, or journals, and edges encode relationships such as citation, co-authorship, or institutional affiliation Newman [2010].

The most common representation is the citation network, where each node represents a manuscript and a directed edge from m' to m indicates that m' cites m Kessler [1963], Small [1973]. This structure underlies a wide range of influence metrics, including citation counts, PageRank-style centrality measures Page et al. [1999], Chen et al. [2007], and various field-normalized impact indicators Waltman et al. [2011], Waltman and van Eck [2015]. Similarly, co-authorship graphs model collaborations by connecting authors who have jointly produced a manuscript, often yielding insights into community structure and knowledge diffusion Newman [2001], Barabási [2002].

While these graph-based approaches have proven useful in many applications, they exhibit two structural limitations (below) that impede conventional scientometrics in accuracy and precision.

- **Boolean edge semantics:** Relationships graphs such as citation and coauthorship networks are typically encoded as boolean edges (present or absent), without capturing the intensity, quality, or context of the interaction. In these cases, authorship contributions and citations implicitly are treated as equal, leading to many distortions and major exploits in downstream scientometrics.